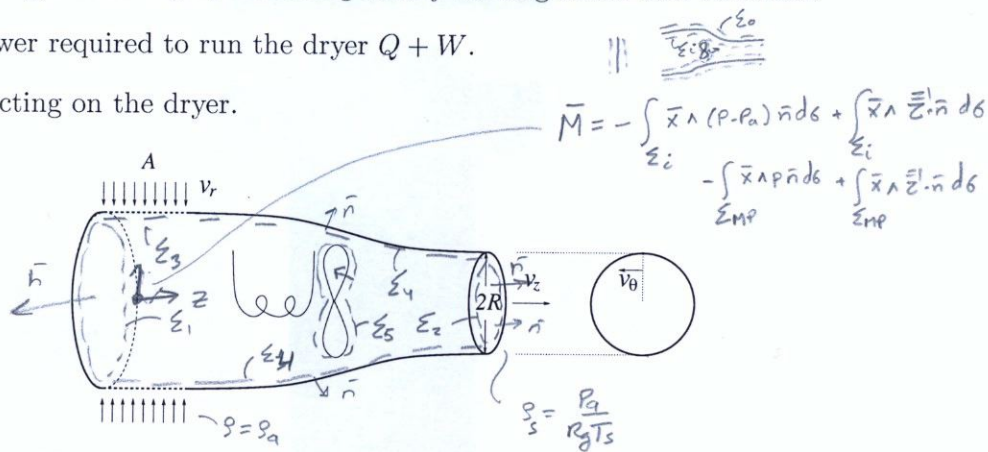


(CL7)

The axisymmetric industrial dryer of the figure is designed to provide a steady hot jet of radius R , mass flux G , and temperature T_s . Air enters the dryer from the ambient, at temperature and pressure T_a and p_a , through a radial inlet of surface A . Inside the dryer there exists an electric resistance that releases an amount of heat Q per unit time, as well as a fan of power W (work per unit time). Because of the presence of the fan, the air velocity at the exit has both azimuthal and axial components, v_θ and v_z , such that $v_\theta/v_z = \tan(\beta)$. In the analysis, assume that the fan and the inner walls of the dryer are thermally insulated, that the air behaves as an ideal gas with constant R_g and specific heat at constant pressure c_p , that gravity has a negligible effect on the fluid motion, and that the flow properties are uniform at the inlet and outlet sections. For given values of G , T_s , T_a , p_a , A , R , R_g , c_p and $\tan(\beta)$, determine:

- The velocity at the entrance section v_r as well as the two velocity components at the exit v_θ and v_z .
- The force acting on the dryer, indicating clearly its magnitude and direction.
- The total power required to run the dryer $Q + W$.
- The torque acting on the dryer.



CONT:

$$G = \rho_a v_r A = \frac{p_a}{R_g T_s} v_z \pi R^2$$

MOMENTUM:

$$\frac{d}{dt} \int_{V_c} \rho \vec{v} dV + \int_{\xi_1 + \xi_2 + \xi_3 + \xi_4 + \xi_5} \rho \vec{v} (\vec{v} - \vec{v}_c) \cdot \vec{n} d\sigma = - \int_{\xi_1 + \xi_2 + \xi_3 + \xi_4 + \xi_5} (p - p_a) \vec{n} d\sigma + \int_{\xi_5} p \vec{n} d\sigma + \int_{\xi_1 + \xi_2 + \xi_3 + \xi_4 + \xi_5} \vec{E}^1 \cdot \vec{n} d\sigma$$

$$\frac{p_a}{R_g T_s} v_z^2 \pi R^2 \vec{e}_z = -\vec{F} \Rightarrow \vec{F} = -\frac{p_a}{R_g T_s} v_z^2 \pi R^2 \vec{e}_z$$

ENERGY:

$$\frac{d}{dt} \int_{V_c} \rho \left(e + \frac{v^2}{2} \right) dV + \int_{\xi_1 + \xi_2 + \xi_3 + \xi_4 + \xi_5} \rho \left(e + \frac{v^2}{2} \right) (\vec{v} - \vec{v}_c) \cdot \vec{n} d\sigma = - \int_{\xi_1 + \xi_2 + \xi_3 + \xi_4 + \xi_5} p \vec{v} \cdot \vec{n} d\sigma + \int_{\xi_1 + \xi_2 + \xi_3 + \xi_4 + \xi_5} \vec{v} \cdot \vec{E}^1 \cdot \vec{n} d\sigma + \int_{\xi_1 + \xi_2 + \xi_3 + \xi_4 + \xi_5} \kappa \nabla T \cdot \vec{n} d\sigma + \int_{V_c} Q dV$$

$$G \left[\left(c_p T_s + \frac{v_\theta^2 + v_z^2}{2} \right) - \left(c_p T_a + \frac{v_r^2}{2} \right) \right] = p_a v_r A - p_a v_z \pi R^2 + W + Q$$

$$W + Q = G \left[\left(c_p T_s + \frac{v_\theta^2 + v_z^2}{2} \right) - \left(c_p T_a + \frac{v_r^2}{2} \right) \right]$$

ANGULAR MOMENTUM:

$$\frac{d}{dt} \int_{V_c} \rho \vec{x} \wedge \rho \vec{v} dV + \int_{\xi_1 + \xi_2 + \xi_3 + \xi_4 + \xi_5} (\vec{x} \wedge \rho \vec{v}) (\vec{v} - \vec{v}_c) \cdot \vec{n} d\sigma = - \int_{\xi_1 + \xi_2 + \xi_3 + \xi_4 + \xi_5} \vec{x} \wedge (p - p_a) \vec{n} d\sigma - \int_{\xi_5} \vec{x} \wedge p \vec{n} d\sigma + \int_{\xi_1 + \xi_2 + \xi_3 + \xi_4 + \xi_5} \vec{x} \wedge \vec{E}^1 \cdot \vec{n} d\sigma$$

$$\rho_s v_z v_\theta z \pi \int_0^R r^2 dr \vec{e}_z = -\vec{M}$$

$$\vec{M} = -\rho_s v_z v_\theta \frac{2\pi R^3}{3} \vec{e}_z = -\frac{2}{3} R G v_\theta \vec{e}_z$$